2.29 Numerical Methods: Newton-Raphson Method

Worked Solutions 1–21

1 a)

$f(x) = x^4 + x^2 - 80$. $f'(x) = 4x^3 + 2x$.

$$x_{n+1} = x_n - \frac{x_n^4 + x_n^2 - 80}{4x_n^3 + 2x_n}$$

$x_1 = 3$. $x_2 = 2.91$ (2 DP).

1 b)

$f(x) = x^3 + 2x^2 - 5x - 4$. $f'(x) = 3x^2 + 4x - 5$.

$$x_{n+1} = x_n - \frac{x_n^3 + 2x_n^2 - 5x_n - 4}{3x_n^2 + 4x_n - 5}$$

$x_1 = 2$. $x_2 = 1.87$ (2 DP).

1 c)

$f(x) = e^x + x - 4$. $f'(x) = e^x + 1$.

$$x_{n+1} = x_n - \frac{e^{x_n} + x_n - 4}{e^{x_n} + 1}$$

$x_1 = 1.5$. $x_2 = 1.14$ (2 DP).

2 a)

$f(x) = 2x^4 + x^3 - 5$. $f'(x) = 8x^3 + 3x^2$.

$$x_{n+1} = x_n - \frac{2x_n^4 + x_n^3 - 5}{8x_n^3 + 3x_n^2}$$

$x_1 = 1.1$. $x_2 = 1.15$ (2 DP).

2 b)

$f(x) = \frac{1}{6}x^6 - 8x - 4$. $f'(x) = x^5 - 8$.

$$x_{n+1} = x_n - \frac{\frac{1}{6}x_n^6 - 8x_n - 4}{8x_n^5 - 8}$$

$x_1 = 2$. $x_2 = 2.39$ (2 DP).

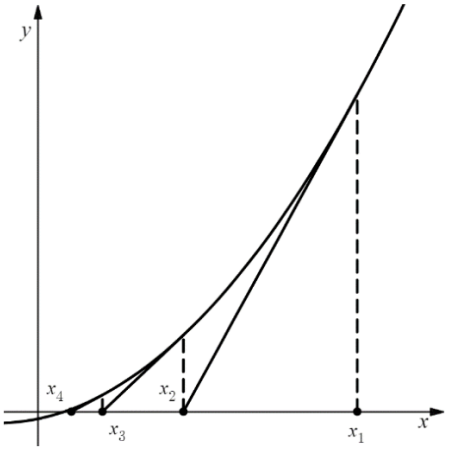
2 c)

$f(x) = e^{2x} - x = 3$. $f'(x) = 2e^{2x} - 1$.

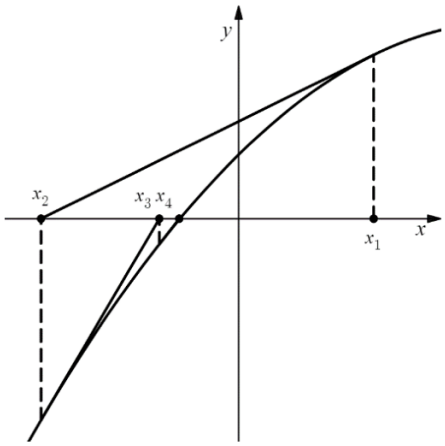
$$x_{n+1} = x_n - \frac{e^{x_n} - x_n - 3}{2e^{x_n} - 1}$$

$x_1 = 0.65$. $x_2 = 1.26$ (2 DP).

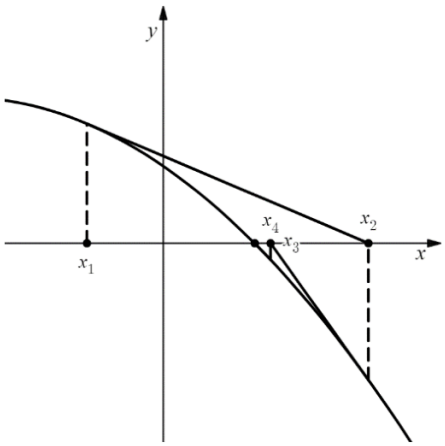
3 a)



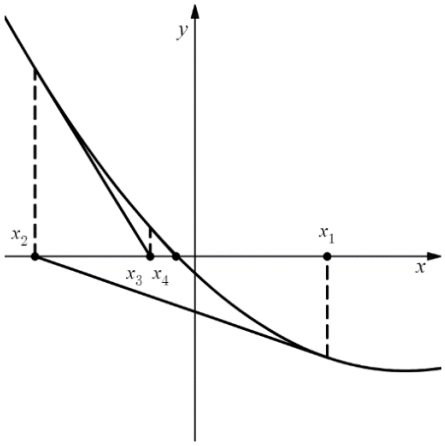
3 b)



3 c)



3 d)



4 a)

It reaches the root at $-\frac{9}{40}$.

4 b)

It coincides with a stationary point, therefore fails to reach a root.

4 c)

It reaches the root at $\frac{89}{40}$.

5 a)

It is already at the root at A.

5 b)

It reaches the root at A.

5 c)

It coincides with a stationary point, therefore fails to reach a root.

5 d)

Closer to B it will reach the root at E, before switching to reach the root at C.

5 e)

It is already at the root at C.

5 f)

It coincides with a stationary point, therefore fails to reach a root.

5 g)

It reaches the root at E.

5 h)

It is already at the root at E.

6)

$f(x) = x^3 - 20$. $f'(x) = 3x^2$.

$$x_{n+1} = x_n - \frac{x_n^3 - 20}{3x_n^2}$$

$$\begin{aligned} x_1 &= 3 \\ x_2 &= 2.740740741\ldots \\ x_3 &= 2.714669625\ldots \\ x_4 &= 2.71441764\ldots \\ x_4 &= 2.714417617\ldots \end{aligned}$$

$\sqrt[3]{20} = 2.714$ (3 DP).

7 a)

$f(x) = 3x^3 + 4x^2 - 3$. $f(0) = 0 + 0 - 3 = -3 < 0$. $f(1) = 3 + 4 - 3 = 4 > 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 0$ and $x = 1$, there must be a root between $x = 0$ and $x = 1$.

7 b)

$f(x) = 3x^3 + 4x^2 - 3$. $f'(x) = 9x^2 + 8x$.

$$\begin{aligned} x - \frac{3x^3 + 4x^2 - 3}{9x^2 + 8x} &= \frac{x(9x^2 + 8x)}{9x^2 + 8x} - \frac{3x^3 + 4x^2 - 3}{9x^2 + 8x} \\ &= \frac{9x^3 + 8x^2 - (3x^3 + 4x^2 - 3)}{9x^2 + 8x} = \frac{6x^3 + 4x^2 + 3}{9x^2 + 8x} \end{aligned}$$

$$x_{n+1} = x_n - \frac{3x_n^3 + 4x_n^2 - 3}{9x_n^2 + 8x_n} = \frac{6x_n^3 + 4x_n^2 + 3}{9x_n^2 + 8x_n} \text{ as req.}$$

7 c)

$x_2 = 0.7647$ (4 DP). $x_3 = 0.7049$ (4 DP).

7 d)

There is a zero on the denominator of the formula. This is caused by a stationary point on the curve at $x = 0$ and $x = -\frac{8}{9}$.

8 a)

$a = 2$, $d = 3$, $n = n$, $S_n = \frac{12}{n}$.

$$\begin{aligned} \frac{12}{n} &= \frac{n}{2}[2 \times 2 - 3(n-1)] \\ &\Rightarrow 24 = n^2[4 - 3n + 3] \\ &\Rightarrow 24 = n^2 - 3n^3 \\ &\Rightarrow 3n^3 - n^2 + 24 = 0 \text{ as req.} \end{aligned}$$

8 b)

$f(n) = 3n^3 + n^2 - 24$. $f'(n) = 9n^2 + 2n$.

$$n_{i+1} = n_i - \frac{3n_i^3 + n_i^2 - 24}{9n_i^2 + 2n_i}$$

$$\begin{aligned} n_1 &= 2 \\ n_2 &= 1.9 \\ n_3 &= 1.894847065\ldots \\ n_4 &= 1.894833765\ldots \end{aligned}$$

Therefore $n = 1.895$ (4 SF).

9 a)

$a = 2$, $r = x$, $n = 8$, $S_8 = 100$.

$$\begin{aligned} 18 &= \frac{2(1-x^8)}{1-x} \\ &\Rightarrow 18(1-x) = 2(1-x^8) \\ &\Rightarrow 18 - 18x = 2 - 2x^8 \\ &\Rightarrow 2x^8 + 16 - 18x = 0 \\ &\Rightarrow x^8 - 9x + 8 = 0 \text{ as req.} \end{aligned}$$

9 b)

$f(x) = x^8 - 9x + 8 = 0$. $f'(x) = 8x^7 - 9$.

$$x_{n+1} = x_n - \frac{x_n^8 - 9x_n + 8}{8x_n^7 - 9}$$

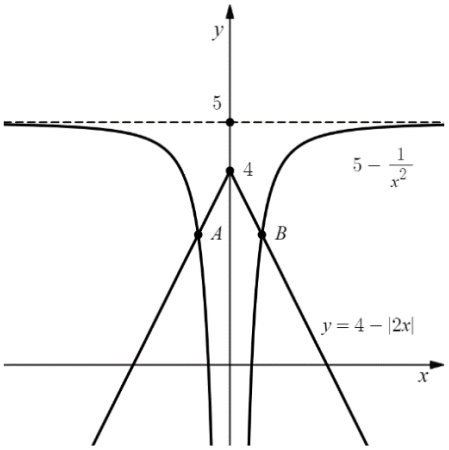
$$\begin{aligned} x_1 &= 0.9 \\ x_2 &= 0.913246\ldots \\ x_3 &= 0.914084\ldots \\ x_4 &= 0.914087\ldots \end{aligned}$$

Therefore $x = 0.9141$ (4 SF).

9 c)

$S_\infty = \frac{2}{1-0.914} = 23.3$ (3 SF).

10 a)



10 b)

$5-\frac{1}{x^2}=4-(-2x). \Rightarrow -\frac{1}{x^2}=2x-1 \Rightarrow 1=-2x^3+x^2. \Rightarrow 2x^3-x^2+1=0$ as req.

10 c)

$f(x)=2x^3-x^2+1.\ f'(x)=6x^2-2x.$

$$\begin{aligned} x-\frac{2x^3-x^2+1}{6x^2-2x} &= \frac{x(6x^2-2x)}{6x^2-2x}-\frac{2x^3-x^2+1}{6x^2-2x} \\ &= \frac{6x^3-2x^2-(2x^3-x^2+1)}{6x^2-2x}=\frac{4x^3-x^2-1}{6x^2-2x} \\ &= \frac{-4x^3+x^2+1}{2x-6x^2} \end{aligned}$$

Therefore $x_{n+1}=\frac{-4x_n^3+x_n^2+1}{2x_n-6x_n^2}$ as req.

10 d)

$$\begin{aligned} x_1 &= -1 \\ x_2 &= -0.75 \\ x_3 &= -0.6666666666\ldots \\ x_4 &= -0.6574074074\ldots \\ x_5 &= -0.6572981213\ldots \\ x_6 &= -0.6572981061\ldots \end{aligned}$$

Therefore $x=-0.657$ (3 DP).

10 e)

$2x-6x^2=0 \Rightarrow x_1=\frac{1}{3}$ or $x_1=0$.

11 a)

$f(x)=x-2+\sin x.\ f(1)=1-2+\sin 1=-0.1585\ldots<0.\ f(2)=2-2+\sin 2=0.9092\ldots>0.$ Since $f(x)$ is continuous and there is a change in sign between $x=1$ and $x=2$, there must be a root between $x=1$ and $x=2$.

11 b)

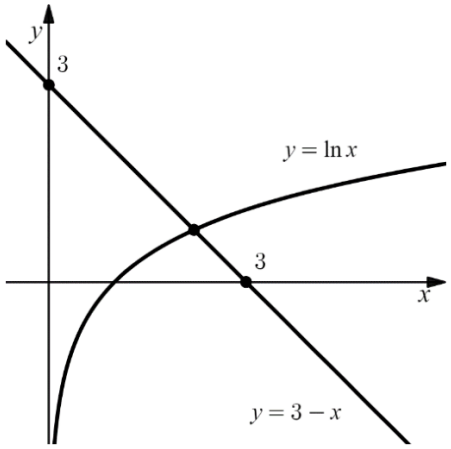
$f(x) = x - 2 + \sin x$. $f'(x) = 1 + \cos x$.

$$x_{n+1} = x_n - \frac{x_n - 2 + \sin x_n}{1 + \cos x_n}$$

$$\begin{aligned} x_1 &= 0.4 \\ x_2 &= 1.030163051\ldots \\ x_3 &= 1.104405861\ldots \\ x_4 &= 1.106059314\ldots \\ x_5 &= 1.106060158\ldots \end{aligned}$$

Therefore $x = 1.11$ (3 SF).

12 a)



12 b)

$f(x) = x - 3 + \ln x$. $f(1) = 1 - 3 + \ln 1 = -2 < 0$. $f(3) = 3 - 3 + \ln 3 = 1.0986\ldots > 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 1$ and $x = 3$, there must be a root between $x = 1$ and $x = 3$.

12 c)

$f(x) = x - 3 + \ln x$. $f'(x) = 1 + \frac{1}{x}$.

$$x_{n+1} = x_n - \frac{x_n - 3 + \ln x_n}{1 + \frac{1}{x_n}}$$

$$\begin{aligned} x_1 &= 2 \\ x_2 &= 2.204568546\ldots \\ x_3 &= 2.207939228\ldots \\ x_4 &= 2.207940032\ldots \end{aligned}$$

Therefore $x = 2.21$ (3 SF).

13 a)

$f(x) = x^2 + \ln(5x)$. $f(0.1) = 0.1^2 + \ln(0.5) = -0.6831\ldots < 0$. $f(0.2) = 0.2^2 + \ln(1) = 0.04 > 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 0.1$ and $x = 0.2$, there must be a root between $x = 0.1$ and $x = 0.2$.

13 b)

$f(x) = x^2 + \ln(5x)$. $f'(x) = 2x + \frac{1}{x}$.

$$\begin{aligned} x - \frac{x^2 + \ln(5x)}{2x + \frac{1}{x}} &= x - \frac{x^3 + x\ln(5x)}{2x^2 + 1} \\ &= \frac{x(2x^2+1)}{2x^2+1} - \frac{x^3 + x\ln(5x)}{2x^2+1} \\ &= \frac{2x^3 + x - (x^3 + x\ln(5x))}{2x^2+1} = \frac{x^3 + x - x\ln(5x)}{2x^2+1} \end{aligned}$$

Therefore $x_{n+1} = \frac{x_n^3 + x_n - x_n\ln(5x_n)}{2x_n^2+1}$ as req.

13 c)

$x_2 = 0.1670$ (4 DP). $x_3 = 0.1911$ (4 DP). $x_4 = 0.1927$ (4 DP).

13 d)

$x^2 \geq 0$, therefore $2x^2 + 1 \geq 1$ so there is no fail case.

14 a)

$\frac{dy}{dx} = 3 - 2x + \frac{10k}{kx+1}$. When $x = 3$, $\frac{dy}{dx} = 0$. Therefore $3 - 6 + \frac{10k}{3k+1} = 0 \Rightarrow \frac{10k}{3k+1} = 3 \Rightarrow 10k = 9k + 3$. $\Rightarrow k = 3$ as req.

14 b)

$f(x) = 3x - x^2 + 10\ln(3x+1)$. $f(7) = 21 - 49 + 10\ln(22) = 2.9104\ldots > 0$. $f(8) = 24 - 64 + 10\ln(25) = -7.8112\ldots < 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 7$ and $x = 8$, there must be a root between $x = 7$ and $x = 8$.

14 c)

$f(x) = 3x - x^2 + 10\ln(3x+1)$. $f'(x) = 3 - 2x + \frac{30}{3x+1}$.

$$x_{n+1} = x_n - \frac{3x_n - x_n^2 + 10\ln(3x_n+1)}{3 - 2x_n + \frac{30}{3x_n+1}}$$

$$\begin{aligned} x_1 &= 7 \\ x_2 &= 7.302025187\ldots \\ x_3 &= 7.292362178\ldots \\ x_4 &= 7.292352309\ldots \end{aligned}$$

Therefore $x = 7.292$ (3 DP).

15 a)

$f(x) = 1 - x\sin x$. $f(1) = 1 - \sin 1 = 0.1585\ldots > 0$. $f(2) = 1 - 2\sin 2 = -0.8186\ldots < 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 1$ and $x = 2$, there must be a root between $x = 1$ and $x = 2$.

15 b)

$f(x) = 1 - x\sin x$. $f'(x) = -x\cos x - \sin x$.

$$x_{n+1} = x_n - \frac{1 - x_n \sin x_n}{-x_n \cos x_n - \sin x_n}$$

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 1.114728672\ldots \\ x_3 &= 1.114157127\ldots \end{aligned}$$

Therefore $x = 1.11$ (3 SF).

16 a)

$f(x) = 2^x x^2 - 3$.

$$\begin{aligned} f'(x) &= 2x \times 2^x + \ln 2 \times 2^x \times x^2 \\ &= 2^x x(x \ln 2 + 2) \end{aligned}$$

By using $x_0 = 0$ she will get a fail case caused by the stationary point of the curve at $(0, -3)$.

16 b)

$$x_{n+1} = x_n - \frac{2^{x_n} x_n^2 - 3}{2^{x_n} x_n (x_n \ln 2 + 2)}$$

$x_1 = 1$. $x_2 = 1.185656396\ldots$. $x_3 = 1.16$ (3 SF).

17 a)

Since $\ln 1 = 0$ the point B has coordinates $(1, 0)$.

17 b)

$$\begin{aligned} \frac{dy}{dx} &= x^3 \times \frac{1}{x} + 3x^2 \ln x = x^2 + 3x^2 \ln x \\ &= x^2(1 + 3\ln x) \\ \frac{d^2y}{dx^2} &= x^2 \times \frac{3}{x} + 2x(1 + 3\ln x) \\ &= 5x + 6x \ln x \text{ as req.} \end{aligned}$$

17 c)

$$x_{n+1} = x_n - \frac{x_n^2(1 + 3\ln x_n)}{5x_n + 6x_n \ln x_n}$$

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 0.8 \\ x_3 &= 0.7277668783\ldots \\ x_4 &= 0.7167853958\ldots \\ x_5 &= 0.7165314456\ldots \\ x_6 &= 0.7165313106\ldots \end{aligned}$$

Therefore $x = 0.717$ (3 SF).

18 a)

$f(x) = x^2e^x - 3x + 1$. $f(0.4) = 0.4^2e^{0.4} - 1.2 + 1 = 0.03869\ldots > 0$. $f(0.5) = 0.5^2e^{0.5} - 1.5 + 1 = -0.87819\ldots < 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 0.4$ and $x = 0.5$, there must be a root between $x = 0.4$ and $x = 0.5$.

18 b)

$f(x) = x^2e^x - 3x + 1$. $f'(x) = x^2e^x + 2xe^x - 3$.

$$\begin{aligned} x - \frac{x^2e^x - 3x + 1}{x^2e^x + 2xe^x - 3} &= \frac{x(x^2e^x + 2xe^x - 3)}{x^2e^x + 2xe^x - 3} - \frac{x^2e^x - 3x + 1}{x^2e^x + 2xe^x - 3} \\ &= \frac{x^3e^x + 2x^2e^x - 3x - (x^2e^x - 3x + 1)}{x^2e^x + 2xe^x - 3} \\ &= \frac{x^3e^x + x^2e^x - 1}{x^2e^x + 2xe^x - 3} \end{aligned}$$

Therefore $x_{n+1} = \frac{x_n^3e^{x_n} + x_n^2e^{x_n} - 1}{x_n^2e^{x_n} + 2x_ne^{x_n} - 3}$ as req.

18 c)

$x_1 = 0.4$. $x_2 = 0.425$ (3 SF). $x_3 = 0.426$ (3 SF).

18 d)

$f(x) = x^2e^x + 2xe^x - 3$.

$$\begin{aligned} f'(x) &= x^2e^x + 2xe^x + 2xe^x + 2e^x - 3 \\ &= e^x(x^2 + 4x + 2) - 3 \end{aligned}$$

$$x_{n+1} = x_n - \frac{x_n^2e^{x_n} + 2x_ne^{x_n} - 3}{e^{x_n}(x_n^2 + 4x_n + 2) - 3}$$

$$\begin{aligned} x_1 &= 0.6 \\ x_2 &= 0.627760752\ldots \\ x_3 &= 0.6131568758\ldots \\ x_4 &= 0.6202326814\ldots \\ x_5 &= 0.6166406397\ldots \\ x_6 &= 0.618418127\ldots \\ x_7 &= 0.6175263781\ldots \end{aligned}$$

Therefore $x = 0.618$ (3 SF).

18 e)

$x \approx 0.62$ is approximately the x-coordinate of the minimum of the curve.

19 a)

Stat pt when $f'(x) = 0$. $f(x) = x \times 3^x - 8x^2 - 2x$. $f'(x) = 3^x + x \times \ln 3 \times 3^x - 16x - 2$. $\Rightarrow 3^x + x \times \ln 3 \times 3^x - 16x - 2 = 0$. $\Rightarrow 3^x + (3^x \ln 3 - 16)x - 2 = 0$ as req.

19 b)

$f'(x) = 3^x + (3^x \ln 3 - 16)x - 2$.

$$\begin{aligned} f''(x) &= \ln 3 \times 3^x + (3^x \ln 3 - 16) + (3^x (\ln 3)^2)x \\ &= 2 \ln 3 \times 3^x - 16 + (\ln 3)^2 \times 3^x x \\ &= (2 \ln 3 + (\ln 3)^2 x)3^x - 16 \end{aligned}$$

$$x_n - \frac{3^{x_n} + (3^{x_n} \ln 3 - 16)x_n - 2}{(2 \ln 3 + (\ln 3)^2 x_n)3^{x_n} - 16} \text{ as req.}$$

19 c)

$$\begin{aligned} x_1 &= 2.2 \\ x_2 &= 2.16472551\ldots \\ x_3 &= 2.1171674945\ldots \\ x_4 &= 2.169950077\ldots \\ x_5 &= 2.17036111\ldots \\ x_6 &= 2.170262147\ldots \end{aligned}$$

$x = 2.17$ (3 SF).

20 a)

$N = 8e^{10-0.25t} \sin(3.15 - 0.19t)$.

$$\begin{aligned} \frac{dN}{dt} &= -0.19 \times 8e^{10-0.25t} \cos(3.15 - 0.19t) - 0.25 \times 8e^{10-0.25t} \sin(3.15 - 0.19t) \\ &= e^{10-0.25t}[-\tfrac{38}{25} \cos(3.15 - 0.19t) - 2 \sin(3.15 - 0.19t)] \end{aligned}$$

Max when $\frac{dN}{dt} = 0$. So when $t \approx 3.4646$, Consider a point before and after: $\left.\frac{dN}{dt}\right|_{t=3.46455} = 0.31\ldots > 0$. $\left.\frac{dN}{dt}\right|_{t=3.46465} = -0.13\ldots < 0$. Therefore given continuity between the two points and the change in sign, true that $t \approx 3.4646$.

20 b)

$$t_{n+1} = t_n - \frac{8e^{10-0.25t_n} \sin(3.15 - 0.19t_n)}{e^{10-0.25t_n}[-\frac{38}{25} \cos(3.15 - 0.19t_n) - 4 \sin(3.15 - 0.19t_n)]}$$

$$\begin{aligned} t_1 &= 16 \\ t_2 &= 16.4503895\ldots \\ t_3 &= 16.57120547\ldots \\ t_4 &= 16.57891752\ldots \end{aligned}$$

16 days and 14 hours.

20 c)

When $t = 0$ there are $N = -1481$ which does not make sense as the initial number of views.

20 d)

The graph is negative after 16 days and 14 hours so the model would no longer be valid.

21 a)

$f(x) = 0.1 + 0.1x^2 \sin(x - 0.5)$. $f(3) = 0.1 + 0.9 \sin(2.5) = 0.6386\ldots > 0$. $f(4) = 0.1 + 1.6 \sin(3.5) = -0.46122\ldots < 0$. Since $f(x)$ is continuous and there is a change in sign between $x = 3$ and $x = 4$, there must be a root between $x = 3$ and $x = 4$.

21 b)

$f(x) = 0.1 + 0.1x^2 \sin(x - 0.5)$. $f'(x) = 0.1x^2 \cos(x - 0.5) + 0.2x \sin(x - 0.5)$.

$$x_{n+1} = x_n - \frac{0.1 + 0.1x_n^2 \sin(x_n - 0.5)}{0.1x_n^2 \cos(x_n - 0.5) + 0.2x_n \sin(x_n - 0.5)}$$

$$\begin{aligned} x_1 &= 3.5 \\ x_2 &= 3.744957443\ldots \\ x_3 &= 3.71459061\ldots \\ x_4 &= 3.714147013\ldots \\ x_5 &= 3.714146917\ldots \end{aligned}$$

Therefore $x = 3.714$ (3 DP).

21 c)

x	0	0.9285	1.857	2.7855	3.714
y	0.1	0.1358	0.4370	0.6860	0.0002

$$\begin{aligned} \text{area} &\approx \tfrac{1}{2} \times 0.9285 \times [0.1 + 0.0002 + 2(0.1358 + 0.4370 + 0.6860)] \\ &\approx 1.215 \end{aligned}$$

Therefore Area ≈ 1.2 units^2.